Event-TimeRAF: Event-Aware Retrieval-Augmented Foundation Models for Explainable Air Quality Forecasting Under Concept Drift

Author Name, *Student Member, IEEE*

***Abstract*—Urban air-quality forecasting is important for public-health warning, environmental management, and city-scale decision support. Recent time-series foundation models have shown promising zero-shot forecasting ability, and TimeRAF improves such models by retrieving relevant time-series sequences from a knowledge base and integrating them with a frozen time-series foundation model. However, air-pollution variation is often influenced by external context, including meteorology, holidays, traffic activity, fire incidents, construction, and other event-driven factors. This draft proposes Event-TimeRAF, an event-aware retrieval-augmented forecasting framework for Dhaka PM2.5 prediction. The proposed methodology combines historical air-quality windows, weather features, calendar context, retrieved similar time-series patterns, retrieved event/news context, concept-drift indicators, and evidence-based explanation generation. Since the implementation has not yet been executed, this manuscript reports the motivation, related work, problem formulation, methodology, and planned experimental setup only. No empirical performance results are claimed in this draft.**

***Index Terms*—Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, TimeRAF, concept drift, explainable forecasting.**

I. Introduction

Air pollution is a major urban risk factor, and fine particulate matter (PM2.5) is especially important because of its association with respiratory and cardiovascular health impacts [1]. Forecasting PM2.5 can support public-health alerts, exposure reduction, transport planning, and environmental policy. In dense cities such as Dhaka, air quality can vary rapidly because of meteorological conditions, traffic patterns, seasonal effects, construction dust, industrial activity, and abnormal events. These characteristics make air-quality forecasting a challenging time-series forecasting problem.

Classical forecasting models and machine-learning models such as ARIMA, tree-based regressors, recurrent neural networks, and Transformer-based models can capture historical patterns when training and testing conditions are similar. However, urban air quality is highly context-dependent. A model trained only on historical numerical trends may underperform during unusual weather, public holidays, traffic disruptions, fire incidents, dust events, or other external shocks. This limitation is closely related to concept drift, where the data-generating process changes over time.

Time-series foundation models (TSFMs) have recently been proposed to improve generalization across forecasting tasks. Models such as TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers (TTM) aim to transfer forecasting knowledge learned from large time-series corpora to new downstream domains [2]–[6]. TimeRAF extends this direction by introducing retrieval augmentation for zero-shot time-series forecasting [7]. It builds a time-series knowledge base, retrieves relevant candidate sequences, and integrates retrieved knowledge using Channel Prompting while keeping the TSFM backbone frozen.

Although TimeRAF is a strong base method, its retrieval knowledge is primarily numerical time-series data. For air-quality forecasting, external context is not optional: rainfall can wash out particulates, low wind speed can increase stagnation, holidays can change traffic patterns, and fire or construction events can introduce abrupt emission changes. Therefore, a natural extension is to retrieve not only similar time-series patterns but also event, weather, and calendar context relevant to the forecast horizon.

This draft proposes Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Dhaka PM2.5 forecasting under concept drift. The planned contributions are:

- an event-aware retrieval-augmented forecasting framework for urban PM2.5 prediction;
- a hybrid knowledge-base design combining air-quality windows, weather summaries, calendar context, and event/news records;
- a TimeRAF-inspired retrieval baseline using random and similarity-based retrieval over historical windows;
- simple concept-drift indicators based on distribution shift, retrieval similarity, weather changes, and event bursts;
- evidence-based explanation generation using retrieved historical cases, event context, weather drivers, drift indicators, and feature importance;
- a planned ablation study comparing no retrieval, random retrieval, time-series retrieval, event retrieval, drift-aware retrieval, and the full Event-TimeRAF framework.

This manuscript is intentionally written as a draft before implementation. Sections on results and discussion contain only the reporting plan and placeholders. No experimental performance values are fabricated.

II. Literature Review and Related Work

A. Time-Series Forecasting

Time-series forecasting has traditionally been addressed using statistical models, machine-learning regressors, and deep

This is a pre-implementation draft prepared for academic supervision. Experimental results are intentionally left as placeholders until the implementation and evaluation are completed.

neural networks. Autoregressive and seasonal models remain useful baselines because they are interpretable and efficient. Gradient-boosted tree models such as XGBoost and LightGBM are widely used for tabular forecasting with lagged and rolling features [8], [9]. Recurrent models such as LSTM and GRU can model sequential dependence [10], [11]. More recently, Transformer-based forecasting models have used patching and channel-independent designs to handle long lookback windows more efficiently [12]. For air-quality forecasting, these approaches are useful but may remain sensitive to distribution shifts and unobserved external events.

B. Time-Series Foundation Models

TSFMs aim to provide transferable forecasting ability across datasets and domains. TimesFM uses a decoder-only architecture for time-series forecasting [2]. Moirai proposes a universal forecasting Transformer trained on a large-scale open time-series archive [3]. MOMENT provides a family of open time-series foundation models for general-purpose time-series tasks [4]. Chronos converts time-series values into tokens and trains language-model architectures for probabilistic forecasting [5]. TTM proposes compact pre-trained models designed for efficient zero/few-shot forecasting [6]. These models motivate the use of pre-trained forecasting backbones, but practical domain adaptation remains difficult when the downstream series is strongly affected by local events and environmental context.

C. Retrieval-Augmented Forecasting

Retrieval-augmented generation was first popularized in NLP by combining parametric models with non-parametric memory [13]. Dense Passage Retrieval (DPR) introduced dense dual-encoder retrieval for open-domain question answering [14]. TimeRAF adapts the retrieval-augmentation idea to zero-shot time-series forecasting [7]. It constructs a time-series knowledge base, retrieves top-k candidate windows using a learnable retriever, and integrates them through Channel Prompting. The base paper reports TTM-Base as the main backbone, context length 512, forecast horizon 96 in most experiments, and $k = 8$ as the default number of retrieved candidates. Its ablations compare random retrieval, cosine retrieval, learnable retrieval, alternative integration mechanisms, candidate counts, knowledge-base choices, and backbone choices.

Event-TimeRAF follows the TimeRAF principle but changes the knowledge source and target domain. Instead of relying only on numerical time-series retrieval, it proposes a hybrid knowledge base containing historical PM2.5 windows, weather summaries, calendar context, and event/news evidence. The first implementation will be TimeRAF-inspired rather than a full reproduction: random retrieval and cosine retrieval will be implemented first, while a dual-encoder MLP retriever and Channel-Prompting-style neural fusion remain advanced extensions.

D. Air-Quality Forecasting and Data Sources

PM2.5 forecasting often uses historical pollutant measurements together with meteorological variables such as temperature, humidity, wind speed, precipitation, and pressure. OpenAQ provides open access to air-quality measurements from monitoring networks [15]. Open-Meteo provides historical and forecast weather APIs suitable for retrieving meteorological covariates [16]. For event context, the GDELT Project provides a large-scale open database of global news and event signals [17]. These sources make it possible to construct a reproducible air-quality forecasting pipeline, subject to data availability for Dhaka.

E. Explainable and Event-Aware Forecasting

Explainability is important for environmental forecasting because users need to understand why a model predicts a rise or fall in pollution. SHAP values provide a model-agnostic framework for estimating feature contributions [18]. However, explanations based only on feature importance may miss temporal and contextual evidence. Event-TimeRAF therefore plans to generate explanations from structured evidence: recent PM2.5 persistence, weather flags, retrieved similar historical windows, retrieved event records, calendar context, and drift indicators. The explanation module will use templates grounded in stored evidence rather than unconstrained language-model generation.

III. Problem Formulation

Let x_t denote the PM2.5 observation at time t. For a forecast time t, the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

where L is the lookback length. Let W_t denote weather covariates aligned with the lookback window, C_t denote calendar features, and E_t denote event context retrieved from an event/news knowledge base. The target is the future PM2.5 sequence

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where H is the forecast horizon.

The forecasting objective is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is retrieved time-series knowledge, R_t^{ev} is retrieved event knowledge, and D_t contains concept-drift indicators. In the initial implementation, the first target setting is $L = 168$ hours and $H = 24$ hours. Longer horizons of 48, 72, and 96 hours are planned after the 24-hour pipeline is stable.

The planned evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

where the earliest 70% of observations are used for training, the next 15% for validation, and the latest 15% for testing. Random splitting is not used because it would leak future temporal information into training.

Table I summarizes the main notation.

TABLE I
MAIN SYMBOLS USED IN THE PROPOSED FORMULATION

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event/news knowledge
D_t	Drift indicators
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

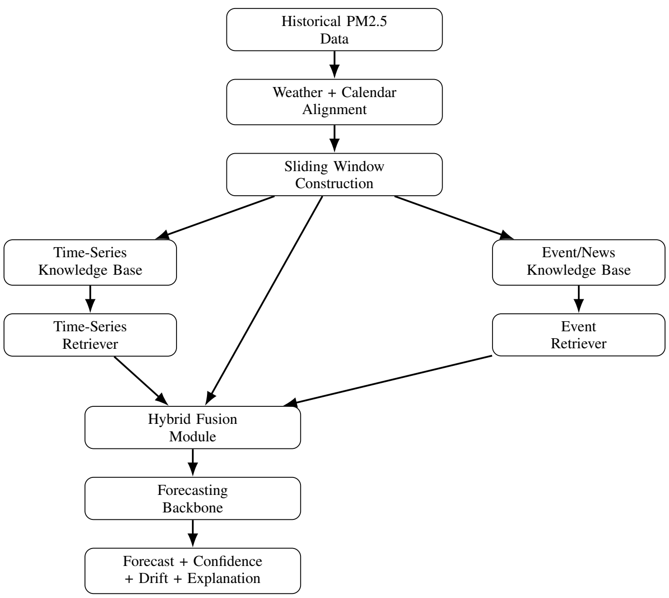

Fig. 1. Planned Event-TimeRAF architecture. The figure is conceptual and does not report experimental results.

IV. METHODOLOGY

A. Overview

Fig. 1 shows the planned Event-TimeRAF pipeline. The system first aligns air-quality observations with weather and calendar features. It then constructs sliding windows and a time-series knowledge base. A time-series retriever returns similar historical windows, while an event retriever returns relevant event/news records around the forecast timestamp. Retrieved evidence is fused with structured features to generate a forecast, confidence proxy, drift score, and explanation.

B. Relationship to TimeRAF

TimeRAF retrieves top-k candidate time-series windows from a curated knowledge base and integrates retrieved candidates using Channel Prompting [7]. It uses a learnable dual-encoder retriever, dot-product similarity, a frozen TSFM backbone, and candidate augmentation during retriever training. Event-TimeRAF preserves the retrieval-augmentation principle but adapts it to Dhaka PM2.5 forecasting and expands the knowledge sources beyond numerical sequences.

The first implementation will not reproduce full TimeRAF. Instead, it will implement a TimeRAF-inspired retrieval baseline using random retrieval and cosine similarity. A dual-encoder MLP retriever, candidate augmentation, and TSFM-based Channel Prompting are reserved for later extensions after the Kaggle-compatible MVP is reproducible.

C. Dataset Construction

The planned air-quality data source is OpenAQ or another accessible source containing Dhaka PM2.5 observations. Timestamps will be converted to Asia/Dhaka time, resampled to hourly frequency if data quality permits, cleaned for impossible values, and interpolated only over short gaps. Weather covariates will be retrieved from Open-Meteo first, with ERA5 considered as a later alternative. Calendar features will include hour, weekday, weekend flag, month, season, public-holiday indicators, Ramadan indicators, and Eid-period indicators where reliable calendar data are available.

All normalization statistics will be fitted on the training split only. The same normalization procedure will be applied to query windows and retrieved candidate windows, following the TimeRAF requirement that input and retrieved sequences use consistent preprocessing.

D. Time-Series Knowledge Base

The time-series knowledge base will contain historical PM2.5 windows. Each record will store a window identifier, start time, end time, input window values, future window values, weather summary, calendar summary, embedding vector or normalized window vector, data split, and normalization identifier. For validation and test evaluation, retrieval will be restricted to training-history windows to prevent leakage.

Given a query window X_t, the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \ldots, r_{t,k}^{ts}\}, \tag{5}$$

where k is the number of retrieved candidates. The default planned setting is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$ and optionally $k = 32$.

E. Event Knowledge Base

The event knowledge base will contain event or news records related to Dhaka or Bangladesh. Planned fields include event identifier, timestamp, location, event category, source, title, summary, tone, and keywords. Event categories include fire, traffic, rainstorm, construction, industrial activity, public gathering, holiday movement, dust, flood, and policy restriction.

For the first implementation, events will be represented as structured aggregate features such as 24-hour event count, 72-hour event count, fire-event count, traffic-event count, industrial-event count, negative-tone average, and top event text. If live event collection is unstable, a cached event dataset will be used and the limitation will be documented.

F. Hybrid Retrieval Score

The planned hybrid retrieval score combines time-series, weather, calendar, and event relevance:

$$s(q,i) = \alpha s_{ts}(q,i) + \beta s_w(q,i) + \gamma s_c(q,i) + \delta s_e(q,i), \quad (6)$$

where s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity, and s_e is event relevance. Initial weights are planned as $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. These weights will be tuned only using validation data after implementation.

G. Fusion and Forecasting Backbone

The MVP fusion method will concatenate PM2.5 lag/rolling features, weather features, calendar features, retrieved time-series summary features, event features, and drift indicators. The first forecasting backbone will be XGBoost or LightGBM because these models are efficient on Kaggle and provide strong tabular baselines. A later extension may use an MLP residual fusion module inspired by Channel Prompting:

$$z_i = \text{Concat}(h_t, r_{t,i}), \quad (7)$$

$$h_t^* = h_t + \frac{1}{k}\sum_{i=1}^{k} \text{MLP}(z_i), \quad (8)$$

where h_t is the input representation and $r_{t,i}$ is the representation of the i-th retrieved candidate. This mirrors the TimeRAF idea of preserving the original input representation while adding uniformly averaged information extracted from retrieved candidates.

H. Concept Drift Detection

Concept drift will be represented using transparent indicators rather than complex online algorithms in the first implementation. Planned indicators include recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst indicator. A simple drift score can be defined as

$$D_t = \lambda_1 z_{\mu,t} + \lambda_2 z_{\sigma,t} + \lambda_3(1 - \bar{s}_{ts,t}) + \lambda_4 b_{event,t}, \quad (9)$$

where $z_{\mu,t}$ and $z_{\sigma,t}$ are normalized mean and variance shifts, $\bar{s}_{ts,t}$ is average retrieval similarity, and $b_{event,t}$ is an event-burst score. The exact thresholds will be selected on validation data.

I. Explanation Generation

The explanation module will generate evidence-based explanations. It will use feature importance or SHAP values, weather flags, retrieved historical windows, retrieved event records, calendar context, forecast direction, and drift score. It will not generate explanations from an unconstrained language model. Each explanation must be traceable to stored evidence.

Table II summarizes the methodological relation between TimeRAF and Event-TimeRAF.

TABLE II
METHODOLOGICAL COMPARISON BETWEEN TIMERAF AND EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF Draft
Target domain	General zero-shot TSF	Dhaka PM2.5 forecasting
Knowledge base	Time-series windows	PM2.5 windows, weather, calendar, events
Retriever	Learnable dual encoder	Random/cosine first; learnable later
Integration	Channel Prompting	Feature fusion first; Channel-Prompting-inspired later
Backbone	Frozen TSFM	XGBoost/LightGBM MVP; TSFM later
Explanations	Retrieved-case visualization	Evidence-based forecast explanation
Drift handling	Not central	Explicit drift indicators

TABLE III
PLANNED DATA SOURCES AND FEATURE GROUPS

Group	Planned Source	Example Features
Air quality	OpenAQ or accessible PM2.5 source	PM2.5, station, timestamp
Weather	Open-Meteo	temperature, humidity, rain, wind, pressure
Calendar	Generated features	hour, weekday, weekend, season, holidays
Events	GDELT or cached event records	event counts, categories, tone, text

V. EXPERIMENTAL SETUP

This section describes the planned experimental setup. The experiments have not yet been executed.

A. Planned Data

Table III lists the planned data sources and feature groups. The final dataset coverage will depend on actual availability of Dhaka PM2.5 measurements and event records.

B. Forecasting Task

The initial task is 24-hour-ahead PM2.5 forecasting with a 168-hour lookback window. The planned split is earliest 70% for training, next 15% for validation, and latest 15% for testing. Random splitting will not be used.

C. Baselines

The planned baselines are:

- persistence baseline;
- seasonal naive baseline;
- XGBoost or LightGBM using PM2.5 lag/rolling features;
- XGBoost or LightGBM using PM2.5, weather, and calendar features;
- random-retrieval baseline;
- cosine TimeRAF-inspired retrieval baseline;
- Event-TimeRAF without drift indicators;
- full Event-TimeRAF with event features and drift indicators.

Optional models such as ARIMA/SARIMA, Prophet, LSTM/GRU, PatchTST, or a TTM wrapper may be added if time and runtime allow.

D. Metrics

The planned metrics are mean squared error (MSE), mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), symmetric MAPE (sMAPE), and R^2. MSE is included for comparability with the TimeRAF base paper, while MAE and RMSE are important for interpretability in air-quality forecasting. Separate event-period, drift-period, and normal-period errors will be reported after drift and event labels are constructed.

E. Ablation Plan

Table IV gives the planned ablation matrix. The entries describe planned model variants only; no result is reported.

Additional planned analyses include candidate-count sensitivity for $k \in \{1, 4, 8, 16\}$, optional $k = 32$, uniform versus similarity-weighted retrieval aggregation, and knowledge-base size sensitivity if runtime allows.

VI. Results and Discussion

No experimental results are available at this draft stage because the data pipeline, retrieval modules, forecasting models, and ablation experiments have not yet been implemented. Therefore, this section intentionally does not report numerical performance, plots, or comparative claims.

After implementation, this section will report:

- main forecasting results across all baselines and Event-TimeRAF variants;
- ablation results for weather, calendar, time-series retrieval, event retrieval, and drift indicators;
- retrieval sensitivity results for different values of k;
- drift-period versus normal-period error;
- event-period case studies;
- example evidence-based explanations;
- computational cost or inference-time comparison if feasible.

The discussion will be written only after real outputs are generated. In particular, the final paper will not claim that Event-TimeRAF improves over baselines unless the saved evaluation tables support that claim.

VII. Conclusion

This draft proposed Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Dhaka PM2.5 forecasting under concept drift. The proposed framework extends the TimeRAF idea by combining time-series retrieval with weather, calendar, event/news context, drift indicators, and evidence-based explanation generation. The first implementation is planned as a Kaggle-compatible MVP using strong tabular baselines and simplified retrieval before moving to advanced TSFM-based fusion.

Because this manuscript is written before implementation, it does not report experimental results or performance claims. The next step is to implement the data pipeline, baselines, retrieval modules, Event-TimeRAF MVP, drift detection, explanation module, and ablation study. The conclusion will be revised after real experimental results are available.

References

[1] World Health Organization, "WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide," 2021. [Online]. Available: https://www.who.int/publications/i/item/9789240034228

[2] A. Das, W. Kong, R. Sen, and Y. Zhou, "A decoder-only foundation model for time-series forecasting," in *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[3] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo, "Unified training of universal time series forecasting transformers," 2024.

[4] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, "MOMENT: A family of open time-series foundation models," 2024.

[5] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," 2024.

[6] V. Ekambaram, A. Jati, P. Dayama, S. Mukherjee, N. H. Nguyen, W. M. Gifford, C. Reddy, and J. Kalagnanam, "Tiny time mixers (TTMs): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series," 2024.

[7] H. Zhang, C. Xu, Y.-F. Zhang, Z. Zhang, L. Wang, and J. Bian, "TimeRAF: Retrieval-augmented foundation model for zero-shot time series forecasting," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 9, pp. 5654–5665, 2025.

[8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[9] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[10] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[11] K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 2014, pp. 1724–1734.

[12] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[13] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W. tau Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[14] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020, pp. 6769–6781.

[15] OpenAQ, "OpenAQ: Open air quality data platform," https://openaq.org/, 2026, accessed: 2026-07-09.

[16] P. Zippenfenig, "Open-meteo.com weather API," 2023. [Online]. Available: https://open-meteo.com/

[17] The GDELT Project, "The GDELT project," https://www.gdeltproject.org/, 2026, accessed: 2026-07-09.

[18] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

TABLE IV
PLANNED ABLATION STUDY

Model	PM2.5	Weather	Calendar	TS Retrieval	Event Retrieval	Drift	Explanation
Persistence	Yes	No	No	No	No	No	No
Seasonal naive	Yes	No	Yes	No	No	No	No
XGBoost-basic	Yes	No	No	No	No	No	No
XGBoost-weather-calendar	Yes	Yes	Yes	No	No	No	No
Random-retrieval baseline	Yes	Yes	Yes	Random	No	No	Limited
Cosine TimeRAF-inspired	Yes	Yes	Yes	Cosine	No	No	Limited
Event-TimeRAF-no-drift	Yes	Yes	Yes	Yes	Yes	No	Yes
Event-TimeRAF-full	Yes	Yes	Yes	Yes	Yes	Yes	Yes